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AI-Powered CV Screening and Shortlisting System

Project Overview

The AI-Powered CV Screening and Shortlisting System is an automated resume evaluation solution designed to assist recruiters and hiring teams in efficiently filtering large volumes of candidate applications. The system leverages Large Language Models (LLMs) through the Groq API to extract key candidate information from PDF resumes, evaluate qualifications against predefined criteria, and automatically shortlist suitable candidates.

Objectives

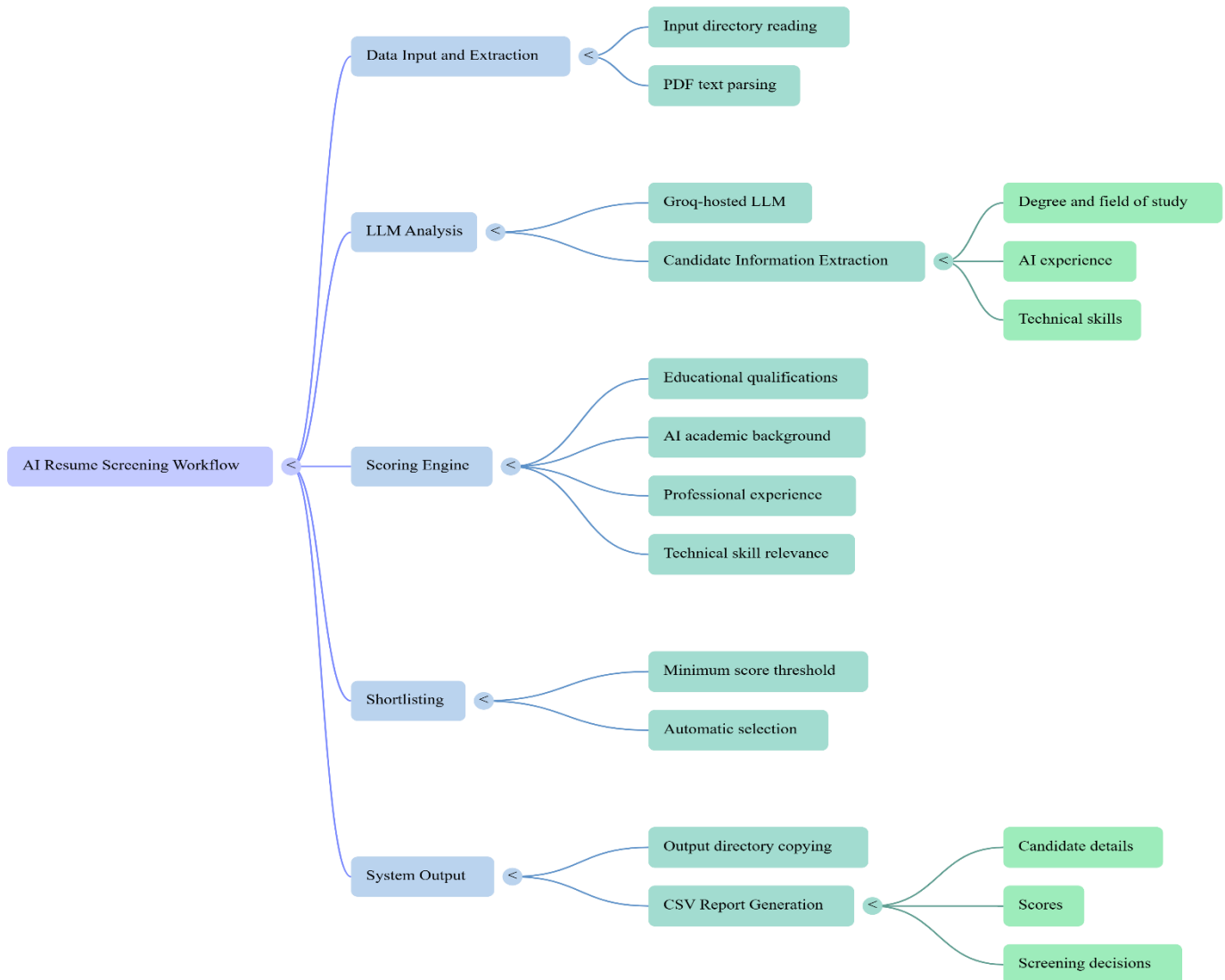
- Automate the resume screening process.
- Reduce manual effort and recruitment time.
- Identify candidates with relevant AI, Machine Learning, and Data Science backgrounds.
- Generate structured candidate reports for easier decision-making.
- Automatically separate shortlisted candidates from non-qualified applicants.

Technologies Used

- Python
- Groq API
- OpenAI-Compatible API Integration
- Pydantic
- Instructor
- PyPDF
- Pandas

System Workflow

1. The system reads PDF resumes from the input directory.
2. Resume text is extracted using PDF parsing techniques.
3. The extracted content is analyzed by a Groq-hosted Large Language Model.
4. Candidate information such as degree, field of study, AI experience, and technical skills is extracted.
5. A scoring engine evaluates each candidate based on educational qualifications, AI-related academic background, professional experience, and relevant technical skills.
6. Candidates meeting the minimum score threshold are automatically shortlisted.
7. Shortlisted resumes are copied to a dedicated output directory.
8. A CSV report containing candidate details, scores, and screening decisions is generated.



Key Features

- Automated PDF resume processing
- AI-powered information extraction
- Candidate scoring and ranking
- Skill-based evaluation
- Automatic shortlisting
- CSV report generation
- Error handling and robust processing
- Scalable architecture for large resume datasets

Benefits

- Significantly reduces recruitment screening time.

- Ensures consistent and objective evaluation criteria.
- Improves hiring efficiency through automated candidate ranking.
- Provides structured candidate data for further analysis.
- Supports high-volume recruitment workflows.

Future Enhancements

- Job Description (JD) matching
- Web-based dashboard
- Google Drive integration
- Email notifications
- Database storage and analytics
- Multi-role recruitment support
- Advanced ATS-style candidate ranking

Conclusion

This project demonstrates the practical application of Artificial Intelligence and Large Language Models in recruitment automation. By combining resume parsing, AI-based information extraction, and intelligent scoring mechanisms, the system streamlines candidate evaluation and helps recruiters quickly identify the most qualified applicants.